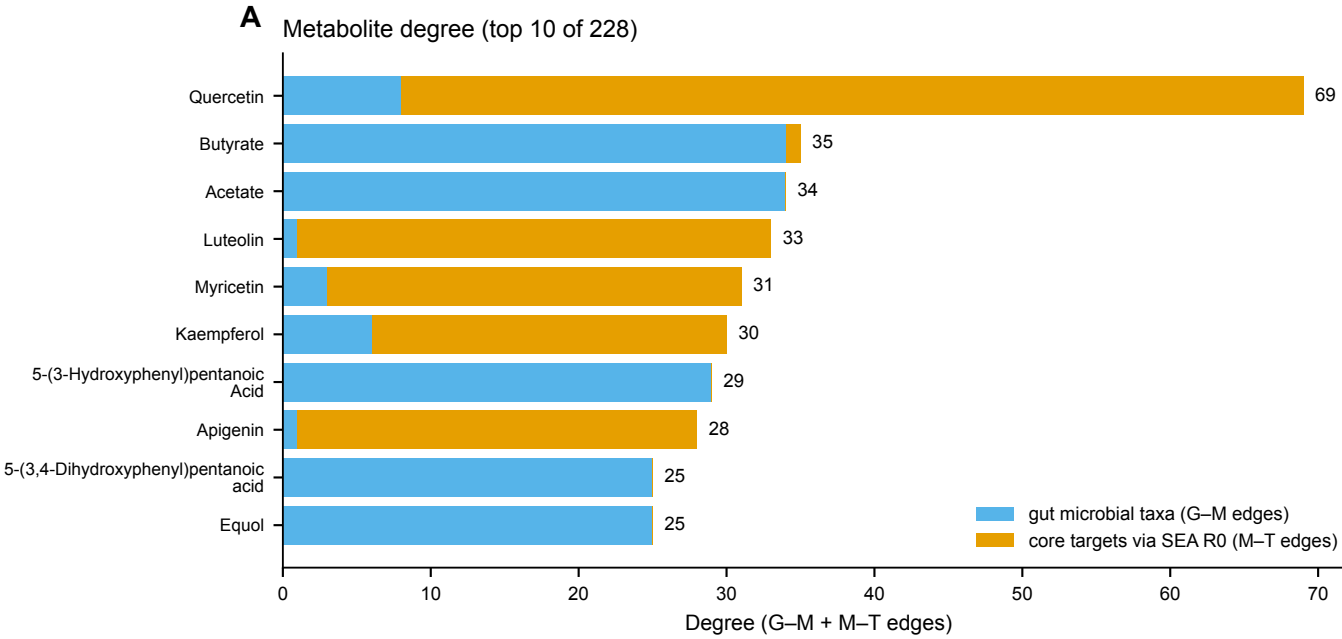
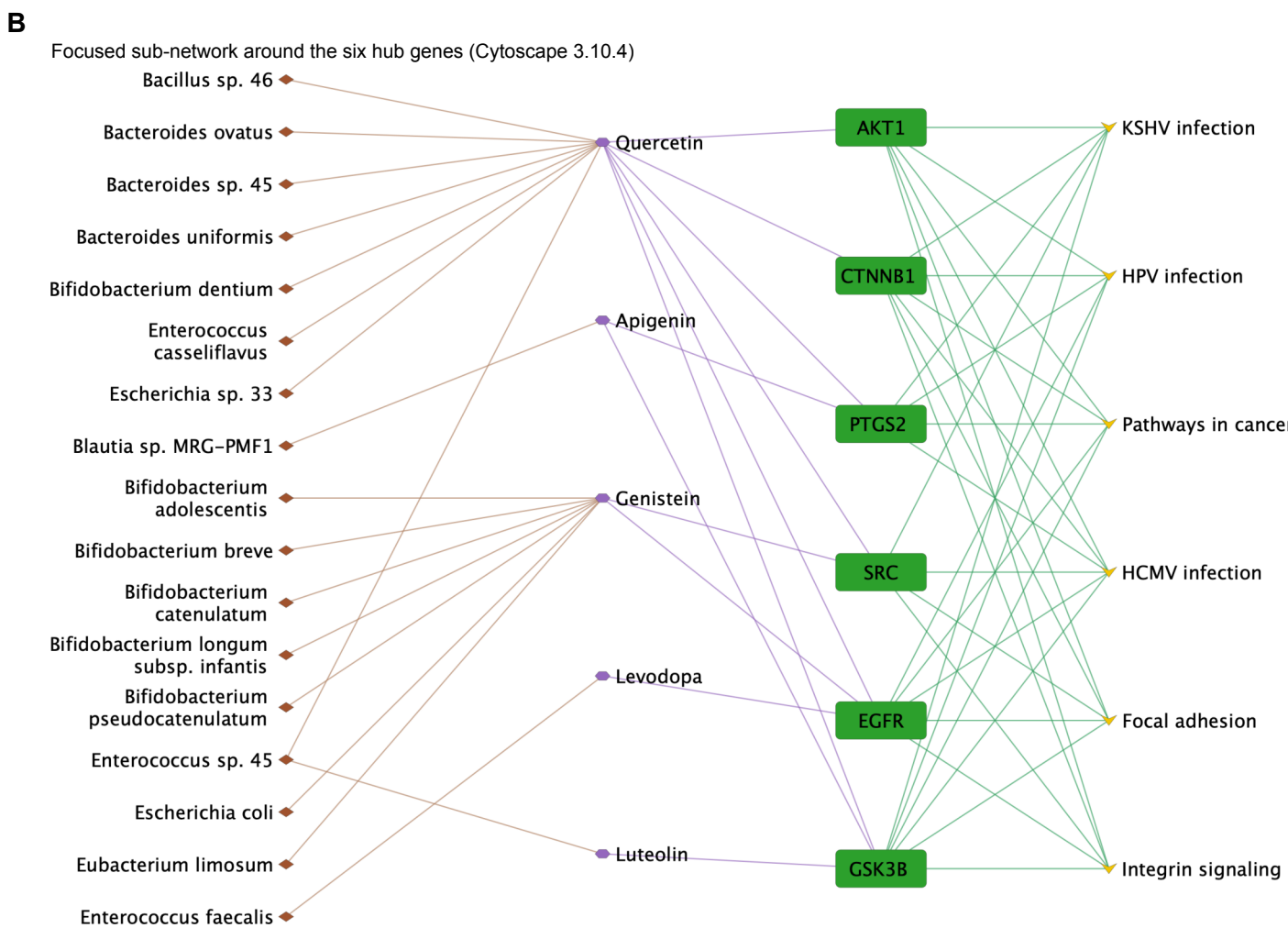




Left to right: gut microbial taxa (17; gutMGene human subset, mixed taxonomic ranks, not audited) → metabolites with a SEA R0 edge to a hub (5) → hub genes (6) → KEGG terms with BH < 0.05 linked to at least 5 hubs (6). Metabolite–target = SEA R0 after removing misattributed records (InChIKey check; potency and concentration not filtered).